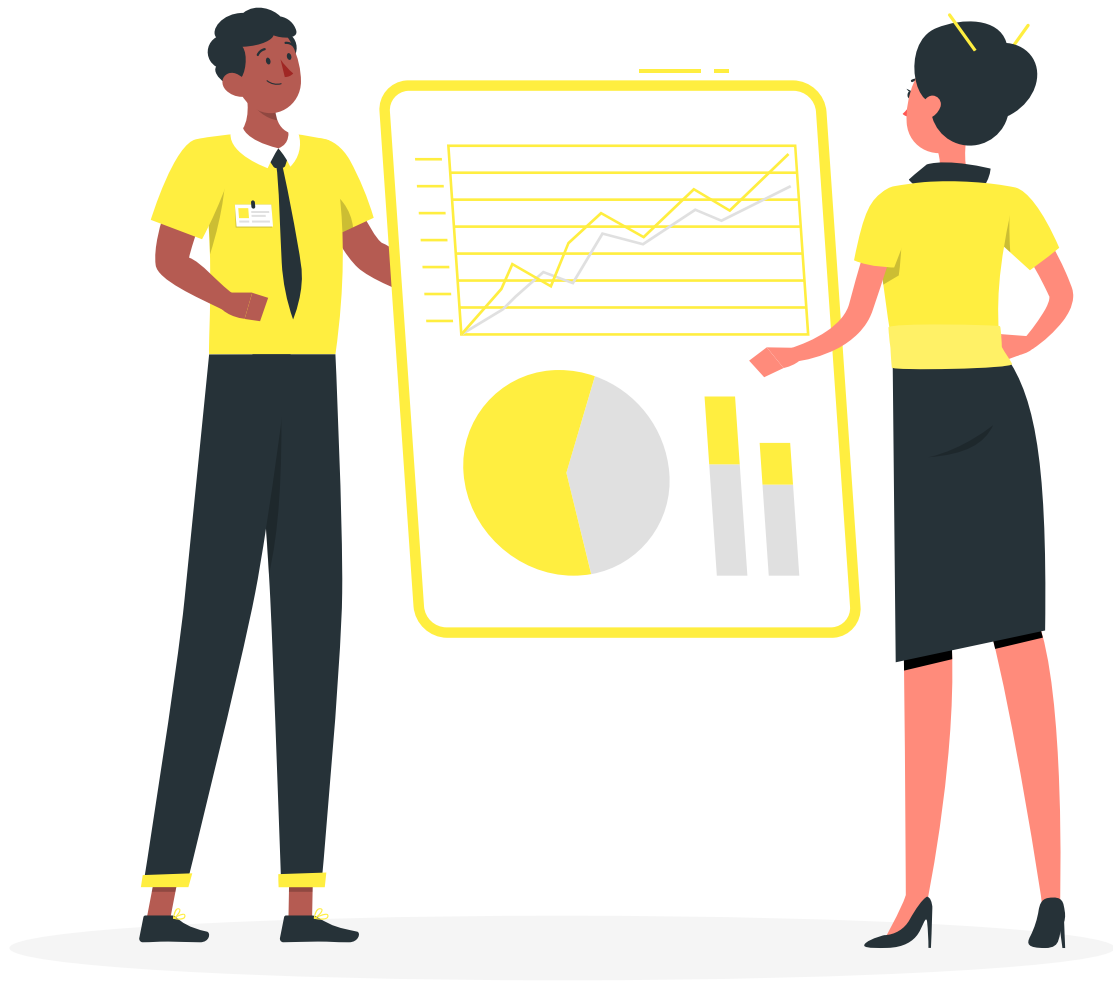




STRATEGIC ANALYSIS OF TELECOM CUSTOMER CHURN

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1. OVERVIEW

In the telecommunications sector, the cost of getting a new customer is outweighs the cost of retaining an existing customers. High churn rates reduce the profit and market share. In this project, Python is used to analyse the data and visualize it and get the reason behind the customer churn in the business

Data Sets

The dataset used for this analysis contains 7,043 rows (representing individual customers) and 21 columns (representing their attributes).

The dataset have below set of information

- Demographic Information
- Services Subscribed
- Account & Financial Information

Tech Stack

Language: Python

Libraries:

- Pandas (Data Wrangling)
- Matplotlib/Seaborn (Visualization)

2. DATA METHODOLOGY & PREPARATION

The analysis is based on two datasets: Customer Data and Zip Code Population Data. Both data sets joined and created the master data set for better analysis

Before we deep dive into analysis, need to cleanup the data and crosschecks the data quality

- Data Integration
- Handling Missing Values
- Duplicate checking

Feature Engineering: There is new column called “Age group” was added based on the customer age and divided into three groups (Young, Adult and Senior)

```
# Upload the dataset

Cust_Data = pd.read_csv(r'D:\01. Portfolio projects\Python Practice\Project 7 (Customer Churn)\telecom_customer_data.csv')
Population_Data = pd.read_csv(r'D:\01. Portfolio projects\Python Practice\Project 7 (Customer Churn)\telecom_zip_code_population_data.csv')

# Merge both dataset

Merged_Data = pd.merge(Cust_Data, Population_Data, how='left', on= 'Zip Code')

# Understand the data

print(Merged_Data.info())
print(Merged_Data.head())
print(Merged_Data.describe())
print(Merged_Data.isnull().sum())
print(Merged_Data.duplicated().sum())

# Handling Missing Values

Merged_Data['Offer'].fillna('None', inplace=True)
Merged_Data['Avg Monthly Long Distance Charges'].fillna(0, inplace=True)
Merged_Data['Avg Monthly GB Download'].fillna(0, inplace=True)
Merged_Data['Multiple Lines'].fillna('No', inplace=True)

Intnet_Col = ['Internet Type', 'Online Security', 'Online Backup', 'Device Protection Plan', 'Premium Tech Support']
for Col in Intnet_Col:
    Merged_Data[Col].fillna('No', inplace=True)
```

3. TARGET VARIABLE ANALYSIS

Overall Churn rate is our target variable. It shows where are at the moment of customer retention. The data was grouped based on the customer status (churned, stayed, joined).

1,869 out of 7,043 customers have left the company. The churn rate is 26.5%.

```
# Target Variable Analysis

tva = Merged_Data.groupby('Customer Status')['Customer ID'].count().reset_index()
tva=tva.rename(columns={'Customer ID': 'No_of_customers'})

tva['Percentage'] = tva['No_of_customers'] / (tva['No_of_customers'].sum()) * 100

print(tva)

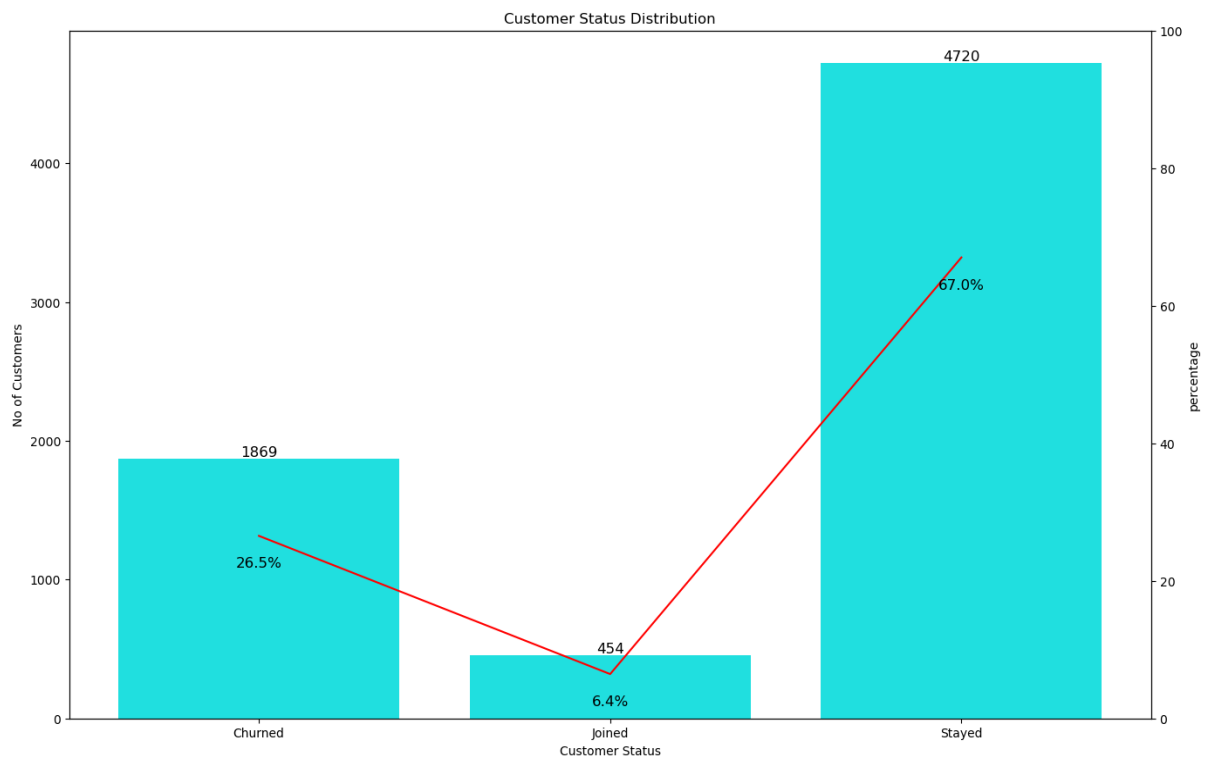
# Target Variable Analysis - Visual

fig, ax1 = plt.subplots(figsize=(16,10))

sns.barplot(data=tva, x='Customer Status', y='No_of_customers', ax=ax1, color = 'cyan')
ax1.set_ylabel('No of Customers')
ax1.set_title('Customer Status Distribution')

ax2=ax1.twinx()
sns.lineplot(data=tva, x='Customer Status', y='Percentage', ax=ax2, color='red')
ax2.set_ylabel('percentage')
ax2.set_ylim(0,100)

for index, row in tva.iterrows():
    ax1.text(index, row['No_of_customers'], f"{row['No_of_customers']}", ha = 'center', va='bottom', fontsize=12)
    ax2.text(index, row['Percentage']-5, f"{row['Percentage']:.1f}%", ha='center', va='bottom', fontsize=12)
```



4. DEMOGRAPHIC ANALYSIS

Customers divided into three different age groups (Young, Adult, and Senior) using `pd.cut`. Churn rate was calculated based on the group and 36.5% of senior customers was discontinued.

This tells we need focus on giving better service for senior customers in terms of offers and support.

```
# Demographic Analysis

Merged_Data['Age_Group'] = pd.cut(Merged_Data['Age'], bins=[0,30,60,100], labels=['Young(0-30)', 'Adult[31-60]', 'Senior(61-100)'])

Age_Churn_rate = Merged_Data.groupby(['Age_Group', 'Customer Status']).size().unstack()

Age_Churn_rate['Churn_rate'] = (Age_Churn_rate['Churned'] / (Age_Churn_rate['Churned'] + Age_Churn_rate['Joined'] + Age_Churn_rate['Stayed']))

Age_Churn_rate_wide = Age_Churn_rate.reset_index()

Age_Churn_rate_long = Age_Churn_rate_wide.melt(id_vars = 'Age_Group',
                                              value_vars = ['Churned', 'Joined', 'Stayed'],
                                              var_name = 'Customer Status',
                                              value_name = 'Count')

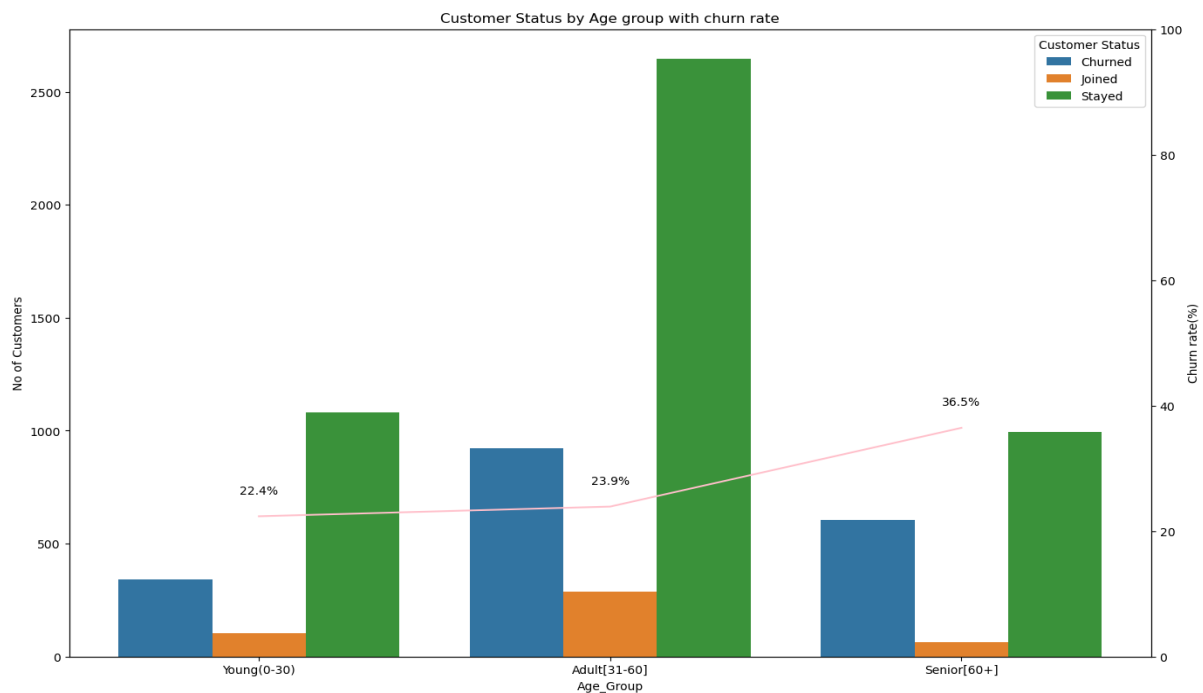
fig, ax1 = plt.subplots(figsize=(16,10))

sns.barplot(data = Age_Churn_rate_long, x= 'Age_Group', y='Count', hue = 'Customer Status', ax=ax1)
ax1.set_ylabel('No of Customers')
ax1.set_title('Customer Status by Age group with churn rate')

ax2= ax1.twinx()
sns.lineplot(data = Age_Churn_rate_wide, x= 'Age_Group', y = 'Churn_rate', color = 'Pink', ax=ax2)
ax2.set_ylabel("Churn rate(%)")
ax2.set_ylim(0,100)

for index, row in Age_Churn_rate_wide.iterrows():
    ax2.text(index, row['Churn_rate']+5, f"{row['Churn_rate']:.1f}%", ha='center', va = 'top')

plt.show()
```



5. SERVICE ANALYSIS

We did two different analysis based on service.

- Internet type(cable, DSL, or fiber option)
- Contract analysis(monthly, one year and two year)

Fiber Optic has an unexpectedly high churn rate of 40.7%, compared to DSL at 18.5%.

Month-to-Month contracts experience massive churn (45.8%)

```
# Service Analysis

svc_analysis = Merged_Data.groupby(['Internet Type', 'Customer Status']).size().unstack()
svc_analysis['Churn_rate'] = (svc_analysis['Churned'] /
                             (svc_analysis['Churned'] + svc_analysis['Joined'] + svc_analysis['Stayed'])) * 100
```

```
In [8]: svc_analysis
Out[8]:
```

Customer Status	Churned	Joined	Stayed	Churn_rate
Internet Type				
Cable	213	56	561	25.662651
DSL	307	115	1230	18.583535
Fiber Optic	1236	101	1698	40.724876
No	113	182	1231	7.404980

```
Contract_analysis = Merged_Data.groupby(['Contract', 'Customer Status']).size().unstack()
Contract_analysis['Churn_rate'] = (Contract_analysis['Churned'] /
                                   (Contract_analysis['Churned'] + Contract_analysis['Joined'] + Contract_analysis['Stayed'])) * 100
```

```
In [9]: Contract_analysis
Out[9]:
```

Customer Status	Churned	Joined	Stayed	Churn_rate
Contract				
Month-to-Month	1655	408	1547	45.844875
One Year	166	24	1360	10.709677
Two Year	48	22	1813	2.549124

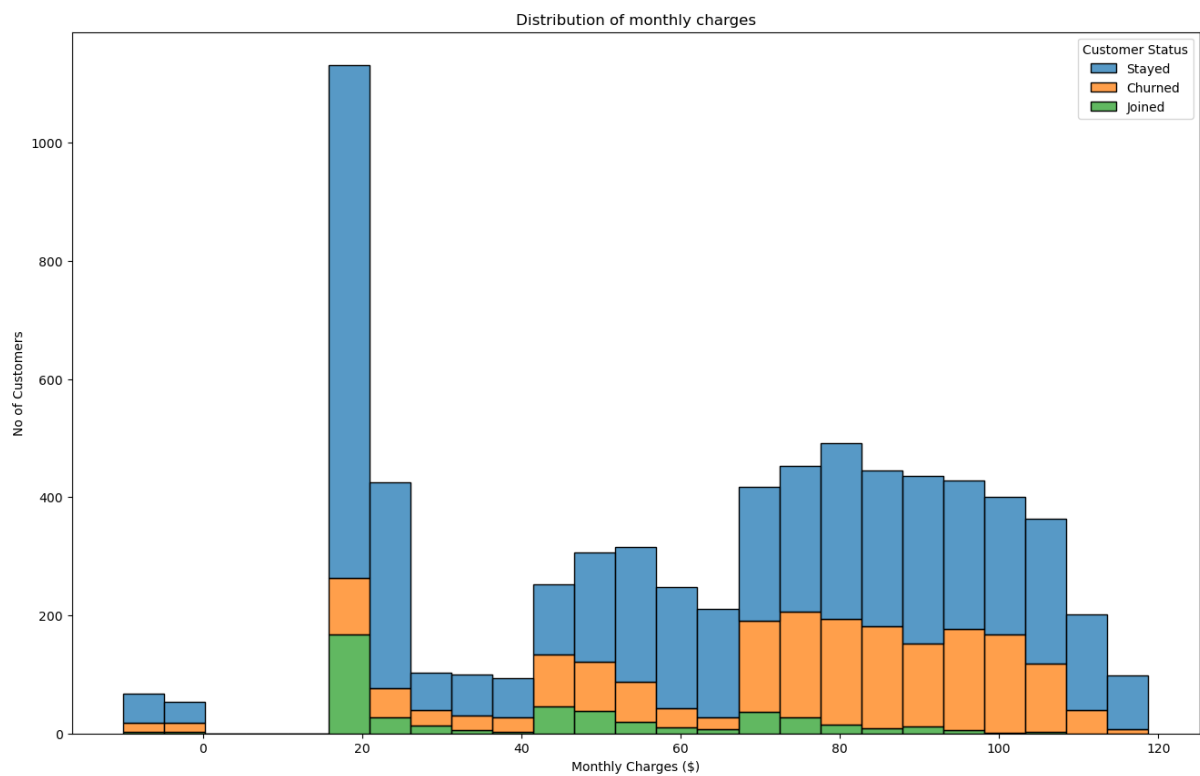
6. FINANCIAL ANALYSIS

In the financial analysis, we have analysed the customer chur rate based on monthly charges.

We have used histogram to see how the distribution between the monthly charges

- Average bill for a retained customer: \$61.73
- Average bill for a churned customer: \$73.34

```
# Financial Analysis
plt.figure(figsize=(16,10))
sns.histplot(Merged_Data, x='Monthly Charge', bins=25, hue='Customer Status', multiple='stack')
plt.xlabel('Monthly Charges ($)')
plt.ylabel('No of Customers')
plt.title('Distribution of monthly charges')
plt.show()
```

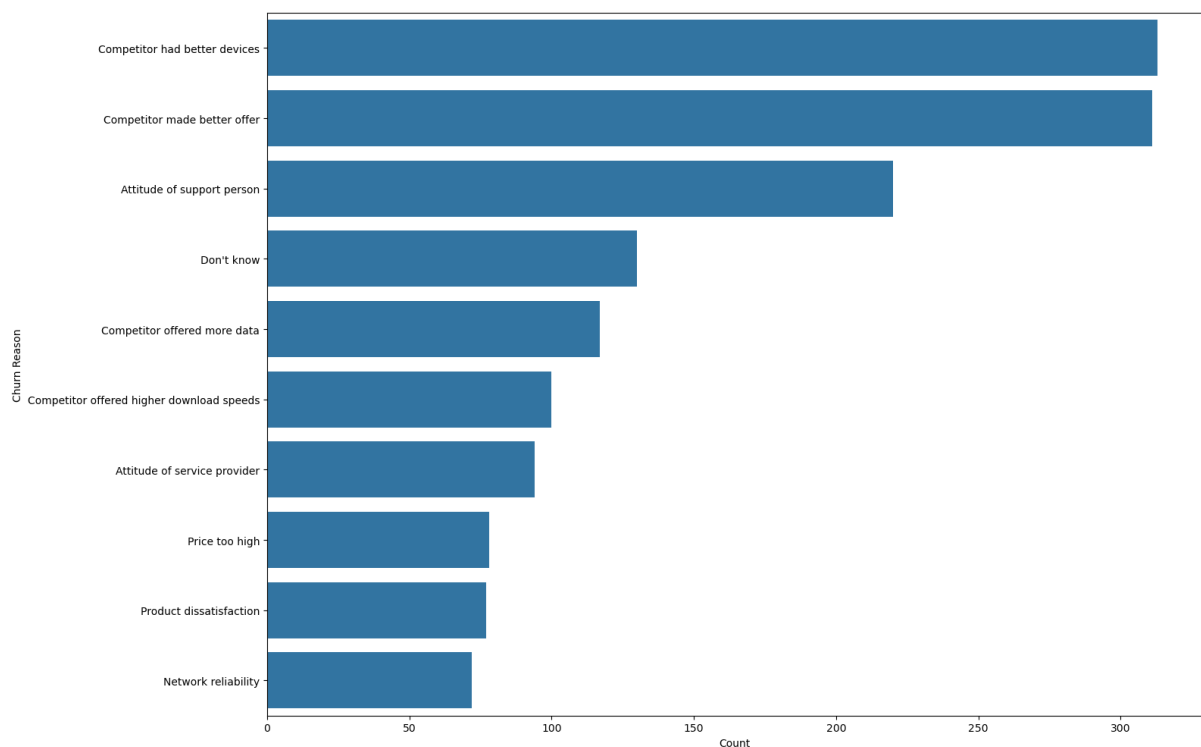


7. ROOT CAUSE ANALYSIS

We have analysed the reasons root Cause for 1,869 churned customers, We able see the top reasons for why customer discontinued

- Better devices (16.7%) ,better offers (16.6%) and Attitude of support person (11.8%) are the top three reasons for leaving from our service

```
Chrun_analysis_1 = Churn_df.groupby('Churn Reason')['Customer ID'].count().  
sort_values(ascending = False).reset_index()  
Chrun_analysis_1.columns = ['Churn Reason', 'Count']  
Chrun_analysis_1['Percentage'] = (Chrun_analysis_1['Count'] / len(Churn_df)) * 100  
  
plt.figure(figsize=(16,10))  
sns.barplot(data=Chrun_analysis_1.head(10), x = 'Count', y = 'Churn Reason')  
plt.tight_layout()  
plt.show()
```

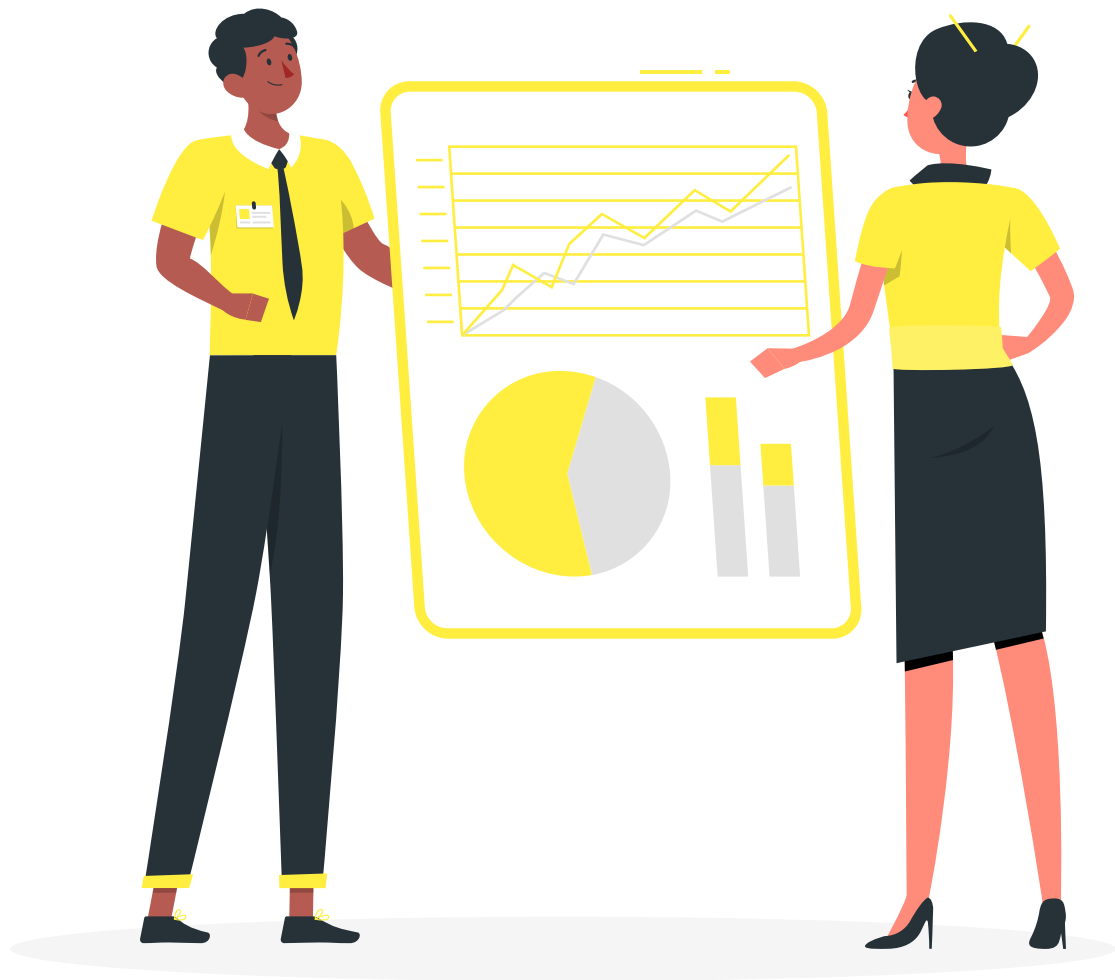


8. CONCLUSION

Based on the above analysis, customer churn is not random and its purely based on the below reason

- Senior customer retention
- Specific products related issues
- Competitor better service

We need to focus on above three to retain the customer and win the business



THANK YOU